Supplementary material for

Specification Surfaces for Incremental Predictive Performance: Estimation, Uncertainty, and Multi-Log Evaluation

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Abstract

This document holds the protocol and its amendments, the proofs, the constructions of the four reporting objects, the correction register, the per-layer and per-fold results, the literature pilot, the diagnosis of the one simulated world where the bootstrap misbehaves, the worked example, the verification harness, the derivation of the coverage calibration, three results the article summarises in a paragraph each — the family-by-encoding crossing, the decision-rule comparison, and the calibration, unseen-category and rolling-origin diagnostics — and the tables the article does not print. The article is self-contained without it. Every number in both documents is generated from the same result files by the same script, so a quantity cannot differ between them.

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S1. Protocol, and the amendments to it

The corpus protocol is `PROTOCOL.md` and the practice pilot's is `AUDIT-PROTOCOL-2.md`; both are in the archive and both were committed before the first result they govern existed. This appendix lists what changed afterwards.

Why amendments are reported rather than folded in. A pre-registration that is quietly edited is worse than none, because it claims a discipline it does not have. Each amendment below names its date, the defect that forced it, and the result it moved.

Amendment 1 — a register's values must recur across time. Registered rule 3.3 admits any attribute reused across two cases on average. On BPI Challenge 2019 that admits the purchasing-document number, which is a document identifier and not a maintained entity: almost none of its values recur across the temporal split. The amendment adds a second, outcome-free condition — at least 50% of test cases must carry a value seen in training — and *both* selections are reported for every log.

Amendment 2 — "any timestamp" had to be implemented. Rule 3.1 excludes timestamps. Matching a fixed list of names did not implement that: it missed five differently-spelled date columns, two of which the registered rules then made the high-cost entity. A column is now a timestamp if its name looks like one or if more than 90% of its non-missing values parse as a date. Both tests read feature values only.

Amendment 3 — a relabelling of the activity is the activity. Some logs carry a second column that is a one-to-one recoding of `concept:name`. Admitting it as a feature admits the activity sequence, which is not available at case creation.

Amendment 4 — coarsenings of the register get their own role. Type, subtype and service are deterministic functions of the item. Sweeping them into the intake block makes the intake block partly a register, and the increment then measures the residual rather than the field. They are a separate role and S4 uses it.

Amendment 5 — the caused-by family is an outcome. A caused-by attribution is the output of diagnosis, not an intake observation. Without this the registered rules made the caused-by configuration item the high-cost entity on the flagship log.

Amendment 6 — the pilot frame's topical filter was nulled. The frame's keyword filter was applied to titles and abstracts, which is a filter on how a paper describes itself rather than on what it does. `results/s06_frame.csv` reports what nulling it does to the estimates.

What was fixed for round nineteen, before any of its results. The design space of Section 3; the reference specification; the exclusion of the intercept-only rung from every admissible set; the six register-quality mechanisms and their levels; the five planned contrasts and the Holm procedure; the two families over which simultaneous bands are taken; and the two-stratum design of the practice pilot, including the decision to adjudicate a sample of the papers the screen *rejected*. The commit that fixed them adds `scripts/spec.py` and `PLAN-ROUND19.md` and no result file.

S2. Proof details and constructions

Proposition 1: the two-point algebra in full

Let a scoring rule take one of two values, with a of the P positives and b of the N negatives in the high block; write $\alpha = a/P$, $\beta = b/N$, $\rho = N/P$ and $\pi = P/(P+N) = 1/(1+\rho)$. Two exact identities follow directly from the definitions in Section 3.5:

$$\mathrm{AUC} = \tfrac{1}{2} + \tfrac{\alpha-\beta}{2}, \qquad \mathrm{AP} = \alpha\,\frac{\alpha}{\alpha+\beta\rho} + (1-\alpha)\pi.$$

The first is the Mann–Whitney statistic with ties at 1/2; the second is the step-wise sum over the two operating points the rule admits. Hence

$$\mathrm{AP} - \pi \;=\; \alpha\left(\frac{\alpha}{\alpha+\beta\rho} - \pi\right).$$

Fix $d = \alpha - \beta > 0$. Every rule on the line $\beta = \alpha - d$ has the same AUC increment $d/2$ over the fully tied baseline, so any two of them give $R_{\mathrm{AUC}} = 0$ exactly. Along that line the AP increment runs from $d(1-\pi)$ at $\alpha = d$, $\beta = 0$

down to $\dfrac{d\rho}{\big(1+(1-d)\rho\big)(1+\rho)}$ at $\alpha = 1$, a ratio of about $1+\rho$. How far apart the two reductions can be pushed therefore grows with the imbalance, which is a property of the construction and not a hidden assumption: refuting the existence of φ needs only that the two be *different*, and at the modest $\rho = 3$ this paper uses they differ by 0.417.

Proposition 2: the cancelling configuration, and what a certificate is

Section 3.5 notes a configuration on which a non-rank-based metric's increment moves under ψ while the ratio does not. The construction is immediate once stated: take $\sigma_{B_1} = \sigma_{B_0}$ and $\sigma_{B_1 \cup f} = \sigma_{B_0 \cup f}$. Then $V(B_1) = V(B_0)$ for every metric and every ψ, so $R = 0$ identically, while V itself moves under ψ whenever m is not rank-based. The configuration is degenerate — the two baselines induce the same scores — and that is why the proposition quantifies over configurations rather than asserting a configuration-by-configuration change.

A *certificate* for an instrument m and a recalibration ψ is a triple of score vectors (a, b, c) on one labelled test set for which

$$\frac{m(b) - m(a)}{m(c) - m(a)} \neq \frac{m(\psi b) - m(\psi a)}{m(\psi c) - m(\psi a)}.$$

By Proposition 2 the existence of one such triple establishes that ψ does not act affinely on m, and therefore that R_m is not invariant — a conclusion about *every* configuration drawn from a finite object. The certificates are searched for by maximising the gap over random triples; `results/s29_certificates.csv` records, for every instrument and every recalibration family, the triple found, the two ratios, and the resulting shift in R. The search reports failure where it fails, and a failure is not evidence of invariance.

A withdrawn computational result

A previous draft reported, as a labelled computational result, a search over a family of synthetic estates for witnesses that no design axis determines another. It is withdrawn and does not appear in this manuscript. Its answer moved between zero and five of twelve ordered pairs as an arbitrary equality tolerance was swept; failing to find a witness is not evidence that one axis determines another; and finding witnesses in a finite synthetic family is not a non-redundancy theorem. Nothing in the manuscript depended on it, which is the clearest argument for removing it rather than repairing it. This is the correction register in the archive.

Proposition 1: the construction

Proof (construction). Take P positives and N negatives and let every score take one of two values, so a scoring rule is described by the pair (a, b) counting positives and negatives in the high block; write $\alpha = a/P$, $\beta = b/N$, $\rho = N/P$. Both baselines are the fully tied rule $\alpha = \beta = 1$, which has AUC $= 1/2$ and AP $= \pi$, the prevalence. Then exactly

$$\mathrm{AUC} - \tfrac{1}{2} = \tfrac{\alpha-\beta}{2}, \qquad \mathrm{AP} - \pi = \alpha\left(\frac{\alpha}{\alpha+\beta\rho} - \pi\right).$$

Fixing $\alpha - \beta$ equal on the two legs makes the AUC increments equal, hence $R_{\mathrm{AUC}} = 0$ exactly — in integers, not to rounding — while sliding α along the admissible range moves the AP increments and sweeps R_{AP}. With $P = 200$, $N = 600$ and therefore $\rho = 3$, the two configurations in `results/s29_prop1.csv` — (α, β) of $(0.8, 0.4)$ and $(0.6, 0.2)$ in one, $(0.9, 0.5)$ and $(0.5, 0.1)$ in the other — have identical R_{AUC} to 2.2×10^{-16} and R_{AP} of -0.250 against -0.667, differing by 0.417. A single φ cannot take one input to two outputs.

The construction is only a proof if those two conditions hold, so `s29_props.py` asserts both: it refuses to write its output if the two configurations fail to share R_{AUC} or fail to separate R_{AP}. A far more imbalanced problem — $\rho = 500$, which this construction previously used — drives the gap close to one, and the modest ρ here shows the separation does not depend on extreme imbalance. □

Proposition 2: the proof

Proof. (ii) ⇒ (i). Each of $V_m(B_0)$ and $V_m(B_1)$ is a difference of two values of m, so an affine change of scale multiplies both by $A(\psi)$ and the additive constant cancels; their ratio, and hence R_m, is unchanged wherever the denominator is nonzero.

(i) ⇒ (ii). Fix ψ and the test set, and write $m(\sigma)$ for $m(\sigma, y)$. For score vectors a, b, c, Assumption 1 admits the configuration $\sigma_{B_0} = a$, $\sigma_{B_0 \cup f} = c$, $\sigma_{B_1} = a$, $\sigma_{B_1 \cup f} = b$, for which $R_m = 1 - \big(m(b) - m(a)\big)/\big(m(c) - m(a)\big)$.

Invariance says

$$\frac{m(b)-m(a)}{m(c)-m(a)} \;=\; \frac{m(\psi b)-m(\psi a)}{m(\psi c)-m(\psi a)} \tag{1}$$

for every admissible triple. First, the map $T : m(\sigma) \mapsto m(\psi\sigma)$ is well defined: if $m(b) = m(b')$ then applying (1) to (a, b, c) and to (a, b', c) with the same a, c gives $m(\psi b) = m(\psi b')$. Second, fix a and c and put $k = \Big(T(m(c)) - T(m(a))\Big)/\Big(m(c) - m(a)\Big)$; then (1) reads $T(u) = T(m(a)) + k\,(u - m(a))$ for every u in the range of m, which is affine with slope $A(\psi) = k$ and intercept $B(\psi) = T(m(a)) - k\,m(a)$. The slope is nonzero because otherwise the transformed denominator vanishes and R_m is undefined rather than invariant. □

Proposition 3: the 3×3 *construction*

Proof (construction). Two levels per axis will not do, and the reason is worth stating: with two levels the two main effects are $|P + Q|$ and $|P - Q|$, where P and Q are the two within-axis differences, so their ordering is decided by $\mathrm{sign}(PQ)$, and a strictly increasing ϕ preserves the sign of each difference. At two levels the ordering is therefore *invariant* and the proposition is false. From three levels a conditional mean averages more than two cells and a convex ϕ reweights them unequally. The 3×3 design in `results/s07_prop3.csv` has $S_A = 0.421$ against $S_B = 0.284$ in its own units and $S_A = 0.232$ against $S_B = 0.390$ under $\phi = \sqrt{\cdot}$, so the leader changes. □

S3. Construction of the inferential objects

Section 4 states what each of the reporting objects is and why it is that and not something else. This appendix carries the constructions themselves, so that the section can be read for the argument and this one for the arithmetic. What is here is the detail a reader would need to reimplement the objects without reading the code.

The multiplier bootstrap for the max-t critical value

Let $\mathcal{F}$ be the declared family, $V_c^{(b)}$ the bth bootstrap draw at cell c, $\bar{V}_c$ the draw mean and $\hat{\sigma}_c$ the per-cell bootstrap standard error. Standardise the draw matrix,

$$Z_{b,c} \;=\; \frac{V_c^{(b)} - \bar{V}_c}{\hat{\sigma}_c}, \qquad b = 1, \ldots, B, \quad c \in \mathcal{F},$$

and let $e_1, \ldots, e_B$ be independent standard normals drawn afresh for each replicate. The *multiplier* statistic is

$$T^{(\mathrm{mult})} \;=\; \max_{c\in\mathcal{F}} \left| B^{-1/2} \textstyle\sum_b e_b Z_{b,c} \right|.$$

Conditionally on the draws, $B^{-1/2} \sum_b e_b Z_{b,\cdot}$ is a centred Gaussian vector whose covariance is exactly the empirical covariance of the standardised draws, so the distribution of $T^{(\mathrm{mult})}$ approximates that of the max-t statistic without any further refitting [1]. The critical value $q_{1-\alpha}$ is the $(1-\alpha)$ quantile of 20,000 such replicates, generated in chunks so that the $20{,}000 \times B$ normal matrix is never held at once.

Two properties are worth stating because they are the reason the construction is used here. The precision of q is governed by the number of multiplier replicates and not by B, so a design whose fitting budget is fixed by the cost of refitting an entire pipeline can still resolve its critical value; and because the multiplier draws are conditional on the observed Z, the construction inherits whatever dependence the moving-block resampling induced across cells, which is what a family-wise statement over a correlated surface requires. The approximation is asymptotic in the same sense the max-t band is: it needs the high-dimensional central limit theorem to hold over a family whose size is comparable to the draw count, which Section 11 states as an assumption rather than a guarantee.

The Monte Carlo interval for q itself is retained. Writing $[q^-, q^+]$ for the order-statistic interval of the multiplier quantile, a cell is called *resolved* only if its band excludes zero at q^+, so residual Monte Carlo noise leaves cells unresolved rather than resolving them in the wrong direction.

Fieller's construction for the reduction

For $\vartheta = V_1/V_0$ with estimates $(\hat{v}_0, \hat{v}_1)$ and bootstrap covariance $(\hat{s}_{00}, \hat{s}_{11}, \hat{s}_{01})$, Fieller's theorem gives the confidence set

$$\Big\{\vartheta : (\hat{v}_1 - \vartheta\hat{v}_0)^2 \le z^2\Big(\hat{s}_{11} - 2\vartheta\hat{s}_{01} + \vartheta^2\hat{s}_{00}\Big)\Big\},$$

whose boundary solves a quadratic in ϑ with leading coefficient $\hat{v}_0^2 - z^2\hat{s}_{00}$. When that coefficient is positive the set is a bounded interval. When it is zero or negative — which is exactly the case in which the denominator $\hat{v}_0$ is not resolvably different from zero at level z — the set is a half-line, the whole line, or the complement of an interval, and reporting a bounded interval for it is reporting a set that is not the confidence set. `results/s02_fieller.csv` records the kind of every set this paper computes, in its `fieller_kind` column.

The three regret designs

Section 4.7 defines the misreport rate $M(s)$ and the regret $G(s)$ and explains why an average of G taken across instruments has no units. The three designs it names are constructed as follows.

A. Metric-specific. For each scalar instrument m separately, the admissible set is restricted to the cells whose metric axis is at m, and $M(s)$ and $G(s)$ are computed over that restriction under the declared measure. Every reported value carries the name of its unit, and no average is taken across instruments. The only cross-instrument statement the design permits is a count: in how many of the five instruments the *direction* of a conclusion is the same, and which ones it is not — and Section 6.8 reports both, because on this corpus two of those counts are four and not five. The design assumes nothing beyond the measure, and it is what the headline uses.

B. Common operational utility. Net benefit at threshold t,

$$\mathrm{NB}(t) \;=\; \frac{\mathrm{TP}(t)}{n} \;-\; \frac{\mathrm{FP}(t)}{n} \cdot \frac{t}{1-t},$$

is measured in true positives per case at an exchange rate the threshold declares, so values at different thresholds are cardinally comparable and may be averaged. The increment is the difference in net benefit between the two arms at the same threshold, computed only on calibrated probabilities (Section 8) because a threshold read as a cost ratio requires one. The resulting quantity is called *decision* regret rather than predictive-performance regret. The design assumes a distribution over thresholds, which is declared and varied rather than fixed.

C. Partial identification. If a reader insists on one number across instruments, the answer depends on a map ϕ from an instrument's units to theirs. Five are declared, in `s23_regret.UTILITY_MAPS`: the *identity*; division by the instrument's remaining *headroom*, the distance left to a perfect score; division by the increment's *standard deviation* across the surface; a signed *square root*; and a logistic squashing of the standardised increment. Each is increasing in the increment and each divides only by positive quantities, so all five fix zero and the *sign* of an increment is invariant to every one of them; what moves is the relative magnitude an instrument contributes to a mixed average. The conclusion is reported as its *range* over the family together with the part invariant to all of it, which is exactly the part the design declines to identify.

Designs A and C are computed on the full declared surface, since neither needs a bootstrap draw. Design B is computed on the calibrated decision-curve cells of Section 8.

Estimating the decomposition

Write $\mathbb{P} = \bigotimes_i w_i$ for the product measure over the axes. Every subset u of axes has a component g_u given by the Möbius sum over the conditional means,

$$g_u(x_u) \;=\; \sum_{v\subseteq u}(-1)^{|u|-|v|}\,\mathbb{E}_{\mathbb{P}}[V \mid x_v]\,, \qquad D_u \;=\; \mathbb{E}_{\mathbb{P}}\Big[g_u^2\Big]\,,$$

and the components are mutually orthogonal, so $\sum_{u\neq\emptyset} D_u = \mathrm{Var}(V)$ exactly. The first-order and total indices of Section 4.5 are $S_i = S_{\{i\}}$ and $S_{T_i} = \sum_{u\ni i} S_u$ for $S_u = D_u/\,\mathrm{Var}(V)$.

The components are computed exactly rather than by a sampling estimator, which the complete factorial makes possible. For that product measure the conditional means $\mathbb{E}_{\mathbb{P}}[V \mid x_v]$ are weighted group means over the cell table, one for each of the 2^k subsets v of the k axes; the Möbius sum then gives every g_u and every $D_u = \mathbb{E}_{\mathbb{P}}[g_u^2]$. The implementation forms the group index once with a mixed-radix encoding of the axis levels and reuses it for all 2^k subsets, so the cost is 2^k passes over the cell table rather than 2^k table joins.

Three checks run on every decomposition and are recorded rather than assumed. The disjoint components must sum to the variance, which is asserted numerically on all 304 decompositions. Duplicating a level with its weight split must leave every component unchanged, and does, to 5.6×10^{-17}. Refining the operating-point grid under the same underlying continuous measure must leave the threshold axis's index unchanged, and moves it by 0.004.

Where the inference surface supplies draws, the whole decomposition is recomputed inside every draw, which gives an interval for every index rather than a point; corpus-level medians are bootstrapped with the log–target pair as the resampling unit, because pairs and not cells are the independent objects. No axis is called dominant anywhere in this paper where those intervals overlap.

S4. Which layer of the register pays

A register has layers. The affected configuration item is the finest; its type, its subtype and the service component it belongs to are deterministic coarsenings, and a buyer choosing how much of a register to populate is choosing between them. The registered rules give coarsenings their own role (S1, amendment 4) so that this question can be asked without the coarsenings contaminating the intake block.

Table S1 reports the increment of each layer over the same baseline, and of the item over each layer. The reading is that the coarse layers are close to free — they are largely recoverable from the intake block — and that what the item adds over its own type is most of what it adds over the intake block. A programme that populates types and subtypes but not items buys little on this task; the finding does not transfer to any other task and is not claimed to.

log	target	level	levels from coarse	cardinality	n	n test	base auc	auc	gain over base
Helpdesk	duration	service_type	0	3	4580	1374	0.629	0.629	0.001
Helpdesk	duration	support_section	1	7	4580	1374	0.629	0.629	0.000
Helpdesk	duration	product	2	21	4580	1374	0.629	0.638	0.009
Helpdesk	duration	product marginal over support_section	-1	21	4580	1374	0.629	0.638	0.009
BPIC19	duration	case:Spend area text	0	20	248445	74534	0.619	0.620	0.000
BPIC19	duration	case:Vendor	1	1944	248445	74534	0.619	0.672	0.052
BPIC19	duration	case:Item Category	2	4	248445	74534	0.619	0.620	0.000
BPIC19	duration	case:Item	3	490	248445	74534	0.619	0.621	0.002
BPIC19	duration	case:Item marginal over case:Item Category	-1	490	248445	74534	0.620	0.621	0.002
BPIC14	handover	CI Type (aff)	0	13	46606	13982	0.641	0.775	0.134
BPIC14	handover	CI Subtype (aff)	1	65	46606	13982	0.641	0.804	0.163
BPIC14	handover	CI Name (aff)	2	3019	46606	13982	0.641	0.898	0.257
BPIC14	handover	CI Name (aff) marginal over CI Subtype (aff)	-1	3019	46606	13982	0.804	0.913	0.109
BPIC14	duration	CI Type (aff)	0	13	46606	13982	0.618	0.662	0.045
BPIC14	duration	CI Subtype (aff)	1	65	46606	13982	0.618	0.662	0.045
BPIC14	duration	CI Name (aff)	2	3019	46606	13982	0.618	0.747	0.129
BPIC14	duration	CI Name (aff) marginal over CI Subtype (aff)	-1	3019	46606	13982	0.662	0.749	0.087

Table S1: The increment of each layer of the register over the same baseline, on the log–target pairs the registered rules of Section 5 admit. 9 rows on excluded pairs are not printed. *Columns.* levels from coarse is the layer's depth, counting from the coarsest layer at 0 inward; the value -1 is not a depth but marks the *marginal* row, which adds the finest layer on top of the second finest. base auc is the AUC of the model this row is measured against — the intake block B_0 with every register layer removed on a layer row, and B_0 together with the second-finest layer on the marginal row. auc is the AUC of that same model with this row's layer added, and gain over base is the difference of the two. Because the marginal row's baseline is not B_0, its gain is not comparable with the rows above it and is not meant to be: it answers what the finest layer is worth once the coarser one is already in the model.

S5. Rolling-origin results, per fold

The single temporal holdout is one split. Table S2 reports the increment at the reference cell under each of the five expanding-origin folds, per log–target pair, so that a reader can see how much of the holdout's answer is the holdout's.

The split axis carries a median 18.9% of the variance in the increment (Table 4) — the largest of the four first-order indices on this corpus, which is worth stating because the split is the axis an evaluation paper is most likely to fix without comment. Reporting a single holdout was a limitation of an earlier version of this work, and where a fold disagrees with the holdout in sign the pair's region label in Table 3 reflects it.

log	target	rolling1	rolling2	rolling3	rolling4	rolling5
BPIC13_closed	handover	-0.019	-0.011	-0.024	-0.012	0.015
BPIC13_incidents	duration	0.001	0.027	0.051	0.030	0.007
BPIC13_incidents	handover	0.030	0.038	0.049	0.032	0.032
BPIC14	duration	0.076	0.100	0.083	0.109	0.112
BPIC14	handover	0.092	0.121	0.152	0.155	0.141
BPIC15_1	duration	0.034	0.030	-0.020	-0.018	0.008
BPIC15_1	handover	0.001	0.010	0.001	0.005	0.004
BPIC15_3	duration	0.166	0.033	0.050	0.031	0.026
BPIC15_3	handover	0.031	0.020	0.037	-0.023	0.007
BPIC15_4	duration	-0.040	-0.004	-0.004	0.031	0.023
BPIC15_4	handover	0.000	-0.018	0.000	-0.032	0.038
BPIC15_5	duration	0.050	0.052	0.024	0.026	0.052
BPIC17	duration	0.003	0.002	0.005	0.010	-0.007
BPIC19	duration	0.044	0.053	0.059	0.018	-0.004
Helpdesk	duration	0.003	-0.004	0.016	-0.039	0.000
RoadFines	duration	0.024	0.010	0.021	0.033	0.041
Sepsis	duration	0.130	0.159	0.154	0.129	0.186
UCI498	duration	-0.007	-0.011	-0.008	-0.006	-0.010
UCI498	handover	0.003	-0.003	0.001	-0.003	-0.005

Table S2: The increment at the reference cell under each rolling-origin fold.

S6. The noisy world, and two explanations that were wrong

In round nineteen the simulation of Section 10 reported that every interval construction fails on the noisy world: coverage 7.0% against a nominal 95%, with the estimator sitting -0.0353 below V_{limit} and a mean interval width of 0.0367. A gap the size of the whole interval is not something a resampling scheme repairs, and the natural write-up was that the estimator is biased in that world and the interval inherits the bias.

That write-up would have been wrong, and the two experiments below are what stopped it. Both were designed so that either could refute the bias reading, and both did.

The gap against sample size. A finite-sample bias closes as n grows. The increment was estimated at six sample sizes from 2,000 to 64,000 in every world, at 30 replicates each, with the penalty held fixed. The sparse world behaves as a finite-sample gap must: $+0.0118$ at $n = 4{,}000$ falling to $+0.0006$

at 64,000. The noisy world does not: $+0.0361$ at $n = 4{,}000$ and still $+0.0329$ at 64,000. The sparse world is the control that makes the noisy result mean something — without it, “the gap did not close” could be a statement about the experiment rather than about the world.

The gap against the penalty. A fixed L_2 penalty is not scale-free, so it can masquerade as bias: it shrinks coefficients materially at $n = 4{,}000$ and barely at $n = 120{,}000$, which would make a large-sample fit a different specification rather than the same one at its limit. Holding n at 4,000 and weakening the penalty by five orders of magnitude moves the noisy world’s gap from $+0.0361$ to $+0.0362$ — that is, not at all.

What was actually wrong. Neither explanation surviving is what sent us to examine the target rather than the estimator, and the target was wrong. The simulation enumerated the generator’s cells, carrying the *true* register value, while in the noisy world the estimator only ever sees a value that is replaced by a uniform draw with probability 20%. Both estimands were therefore computed on a population the estimator does not sample from. Marginalising the noise exactly (Section 10) moves V_{limit} from $+0.1135$ to $+0.0804$. The estimator’s mean was $+0.0781$ throughout. There was no bias; there was a wrong estimand in our own simulation, and it took two arms of evidence against everything else to find it.

The same two experiments, after the correction. Both arms were re-run against the corrected target, and there is nothing left for them to explain: the noisy world’s gap is $+0.0025$ at $n = 4{,}000$ and -0.0007 at 64,000. The sparse world’s gap is unchanged in both runs, because the correction touches only the noisy world, which is what makes it a control rather than a coincidence. Coverage on the noisy world is 94.8%. `scripts/s18_bias_scaling.py --legacy-target` regenerates the pre-correction figures above, so the evidence for this appendix is reproducible rather than quoted from a log we happen to have kept.

The general lesson is the one this paper makes elsewhere about the design space: a quantity is defined relative to a declared population, and a simulation’s population is a declaration like any other. A quality mechanism that acts on the data — which is exactly what the register-quality axis of Section 3 does on real logs — changes what the estimand is, not merely how well it is estimated. The register-quality results of Section 7.4 are reported

world	fixed-model, basic	fixed-model, percentile	nested, basic	nested, bias-corrected	nested, percentile	MC half-width
drift	0.935	0.915	0.910	0.915	0.975	0.030
imbalanced	0.910	0.910	0.935	0.950	0.940	0.030
linear	0.920	0.900	0.915	0.915	0.945	0.030
noisy	0.930	0.930	0.920	0.925	0.920	0.030
nonlinear	0.935	0.925	0.935	0.940	0.950	0.030
sparse	0.660	0.635	0.925	0.675	0.420	0.030

Table S3: Simulation: coverage of the estimator's own limit at the nominal 95%, by world, for the fixed-model and the nested bootstrap under the percentile, basic and bias-corrected constructions. The nested percentile interval is the one that fails, on the sparse world, and the basic construction is the repair; the bias-corrected percentile is not. The last column is the Monte Carlo half-width of every coverage in the row at this replicate count, so a reader can see which differences the simulation resolves. The oracle rows are in results/s10_coverage.csv.

against the degraded register for that reason, and the correction register in the archive records this as the round's largest error of its class.

The two estimands, and the marginalisation

Two estimands are separated, because under misspecification they are not the same. V_{oracle} is the increment between the two Bayes-optimal scoring rules — what the register is worth in the world. V_{limit} is the increment between the two fitted pipelines' limits — what the estimator estimates. A resampling interval can only cover the second, and reporting its coverage of the first as though it were the estimator's coverage is precisely the error the earlier simulation made. Both are reported in Table S3, which is the 200-replicate run this diagnosis was made on. Table S30 is the 1000-replicate run every coverage number in the main text comes from; the two agree on every qualitative statement in this appendix, and the diagnosis is reported against the run that produced it rather than restated against a later one.

The truth has to be the truth the estimator sees. Both estimands are computed by enumerating cells, and in the noisy world the cells the generator writes down are not the cells the estimator observes: a register value is replaced by a uniform draw with probability 20%, so the observed cell (b, g, j) is a mixture over the true ones, with mass and outcome rate

$$\begin{aligned} w(b,g,j) \;&=\; \sum_f \Pr(b,g,f)\;\Pr(j \mid f), \\ \Pr(y{=}1 \mid b,g,j) \;&=\; \frac{1}{w(b,g,j)} \sum_f \Pr(b,g,f)\;\Pr(j \mid f)\, p(b,g,f), \end{aligned}$$

where $\Pr(j \mid f) = (1-r)\mathbf{1}\{j = f\} + r/K$. Enumerating the *true* cells instead scores the estimator against a population it never samples from, and the first version of this simulation did exactly that: it reported a coverage of 7.0% on the noisy world and a bias of -0.0353 that did not exist. S6 gives the two experiments that refuted every finite-sample explanation for that apparent bias and thereby sent us to look at the target. The marginalisation is checked against a Monte Carlo sample, on standardised residuals, at the start of every simulation run.

S7. The worked example on UCI Adult

The standard is not about event logs and not about configuration databases, and the quickest way to show that is to run it on something else. UCI Adult (Census Income) predicts whether a person's income exceeds a threshold from 30,162 records. Suppose an analyst wants to report what `occupation` is worth. It is the expensive field: a survey has to ask about it and code it, where age, sex, race and country come with the sampling frame and education comes with an enrolment record.

Worth compared with what? Against the frame alone, against the frame plus education, or against the frame plus education plus hours worked. Measured how, at which operating point, and with the register how complete. Over the resulting 480 cells the reduction runs 0.200 to 0.947. The baseline axis and the metric axis carry the same first-order share of the variance in the increment to three figures — 25.9% and 25.9% — against 7.2% for the register population, and a one-number report misstates the sign for a mean 24.9% of the other cells. That the two lead axes tie is a property of this dataset and not a general one; it is the sort of thing a surface reports and a single number cannot.

`examples/worked_example.py` produces all of it in 121 statements against the package's public interface, and ends by calling `float()` on the surface so a reader can see what the refusal looks like.

S8. The verification harness

Every number in the manuscript is a macro, defined in the generated file `paper/numbers.tex` that `scripts/make_numbers.py` writes from `results/*.csv`. The manuscript contains no numeric literal outside that file and the generated tables. A number therefore cannot disagree between the abstract, a table and the conclusion: there is one of it.

That removes one class of defect and creates another — a wrong generator makes the whole manuscript wrong *consistently*, and nothing looks odd — so the architecture comes with seven controls.

The generator. `make_numbers.py` emits a visible **??** for any macro whose source file is absent, and `--strict` exits non-zero if one remains. A partial build is therefore visibly partial and cannot silently drop a claim.

The verifier. `scripts/verify_numbers.py` re-derives each checked macro from the result file with code written against the *definition* rather than against the generator, and compares. It also enforces 25 consistency conditions that no single macro expresses — relations that must hold between numbers even when each of them is individually correct. Among them: that every register-quality mechanism passed its executed train-only check; that no surface is labelled uniformly beneficial while carrying an unresolved cell, nor conditionally harmful while carrying a beneficial one; that every region label is one of the six the manuscript defines and that the labels partition the pairs; that Proposition 1's sweep still holds the AUC reduction at zero; that the disjoint functional-ANOVA components sum to one on every decomposition; that no planned contrast reports a p-value the draw count cannot produce, and that its Holm adjustment reproduces from the raw values; that no rank-based instrument's reduction moves under a monotone recalibration, and that every certificate for a non-rank-based one exhibits two ratios that actually differ; that the cells a region label ranges over are exactly the cells the simultaneous family that licenses it controls, and that every decision rule is compared over the declared regret population rather than over whatever the file happened to hold; that the case study's split partitions its cohort; and that the adjudicated set of the practice pilot is exactly the screened-in set. The partition condition is the one that would have caught round nineteen's fifth region label, which the analysis emitted and the manuscript did not define.

The corruption suite. `scripts/s13_attack_numbers.py` is the verifier's own regression suite. It corrupts one thing at a time — a value in a result file, a value in `numbers.tex`, a deleted macro, a train-only check flipped, a region label made inconsistent with its counts, an adjudication file made inconsistent with the coding file — and requires the verifier to notice. Every corruption is applied to a copy in a temporary directory; the suite refuses to start if a previous run left its lock behind, because in an earlier round a corruption suite was killed mid-flight and a corrupted manuscript reached a commit.

A corruption of a *result* file is checked by regenerating the macros from the corrupted copy and then verifying, which is the only version of that test that means anything: without the regeneration the suite asks the verifier about a `numbers.tex` built from clean results, so any quantity the verifier re-derives from a different file than the generator reads is untouched and the corruption passes. One corruption did exactly that, silently, for a round.

The condition on the set, not on the number. Every check above compares one macro with one source file, and a whole class of defect passes all of them: two macros, each perfectly consistent with its own source, that the prose calls by the same name. That is not hypothetical. It is how a case study's increment came to be printed with two values and two signs, and no check in this repository could have seen it.

The repair is a declaration rather than a computation. The verifier `round21_verify.py` carries an ESTIMANDS table: one entry per quantity a reader would name in one phrase, listing the macros that may hold it. Two members of an entry that disagree is a failure; a quantity that legitimately takes two values under two specifications gets two entries with different names, and the manuscript must then use those names. Corruption `G1` in the suite is the test: it moves a value in one result file and lets both the generator and the verifier read the moved value, so every macro-against-source check passes and only the condition on the set fails. **Write the check on the name, not on the number.**

The provenance record. `results/provenance.json` holds the SHA-256 of every analysis script at the moment its outputs were last accepted, with a note saying why, and `make_numbers.py --strict` refuses to certify a manuscript whose numbers come from a script that has changed since. This exists because `--strict` used to pass on a build reading a two-replicate smoke test that had been run under the wrong seed and against an estimand corrected an hour later: every macro resolved cleanly, to a wrong number. A missing file is loud, and a stale one is silent. Modification time cannot answer the question — changing a script's output format does not make its numbers wrong — so acceptance is a deliberate act with a recorded reason.

The compliance linter. `scripts/texlint.py` carries 15 checks: the abstract length, the keyword count, the highlight lengths against their own stated counts, the presence of all five required statements, the absence of any numeric literal in the prose, the absence of a declared set of rhetorical sentence-openers, that every `\input` target exists, that a spelled-out count matches the list it introduces — whether that list is a run of identifiers or a `description` environment — and that no manuscript part, generated table or analysis script carries a stray control character.

Seven of them exist because each caught a defect that had already reached a compiled PDF of *this* manuscript: a spelled-out count that did not match the list it introduced; the word *Appendix* written before a reference that already supplies it, on every appendix reference at once; a macro swallowing the space after it and printing `42.0%--- and`; a macro carrying words quoted inside math mode and printing `18percentagepoints`; a sentence's tail left behind by a scripted edit and printed twice; a *backslash escape interpreted by a scripted edit*, which had written `\ref{x}` into the file as a carriage return followed by `ef{x}` and `\nMacro` as a newline followed by `nMacro`; and a spelled-out count introducing a `description` list of a different length — which is the first defect in the form the first check could not see, and which was in the paragraph above this one, announcing five controls before listing six. Those print as garbage or as a wrong word, produce no wrong number, and every checker in this repository passed on the build that carried them. Each signature is exact — a carriage return that is not a line ending, a line beginning with the tail of a control word, a numeral word whose list is a different length — so each is now a check rather than a thing to remember. All seven were found by *reading the rendered pages*, which is the check no file in this repository replaces.

The claim registry. `scripts/claim_registry.py` writes one row per

> macro: its value, every manuscript location that uses it, the generator file and *line* that defined it — captured from the calling frame, so it cannot drift — the result file that line reads, and whether the verifier re-derives it independently. Because the manuscript contains no numeric literals, the registry is exhaustive by construction: any number a reader can see is a macro, and every macro is a row. `--check` fails a build in which a macro the manuscript uses is unresolved.
>
> The registry also answers the question a reader should ask of the paragraph above: *how many* macros are independently re-derived, not merely how many could be. Its summary row counts them, separately for the *headline* macros — the ones the abstract, the introduction and the conclusion print — and the count is deliberately not quoted here, because it is a property of a build and belongs in the file rather than in a sentence. It is not all of them, and the ones it is not are visible in the same file. Writing the missing re-derivations is also how the largest arithmetic error in the register was found: a range taken over four rows of a table, three of which are corpus medians and one of which is a single pair.

What the harness does not do. It guards numbers thoroughly and prose only where a guard was written by hand. The correction register in the archive groups this project's own errors by class, and its largest class is sentences in which every printed number was correct and the relation asserted between them was not. The generated-macro discipline makes one sub-class of that structurally impossible — the same quantity printed differently in two places — and does nothing about the rest. Holes have been found in this project's checkers in every round including this one — a corruption that passed because nothing regenerated between it and the check, a strict mode that certified a manuscript built from a superseded run, a compliance rule that reads digits and not words — and every one of them was found by a corruption suite, by a provenance record, or by reading the output, never by the checker itself.

The clearest limit is Class G of the register: a quantity computed correctly, printed correctly, and consistent everywhere it appears, while the *definition* it refers to is not the one the estimator addresses (S6). Every guard described above passes such a manuscript, and would have passed it silently. What caught it was constructing two experiments that could each have confirmed the comfortable reading and having both refuse to. No checker in this repository would have done that on its own.

S9. The coverage calibration

Section 6.4 reports every region label and every robustness index under a critical value calibrated at the pair's own ratio of register cardinality to training row count. This section gives the construction the article summarises.

S9.1. The inflation factor has a closed form

Write the basic interval of Section 4.1 as $[\hat{V} - L,\, \hat{V} + U]$. Inflated by a factor c it becomes $[\hat{V} - cL,\, \hat{V} + cU]$, and it covers the truth ϑ exactly when

$$c \;\geq\; \max\left\{\frac{\hat{V} - \vartheta}{L},\; \frac{\vartheta - \hat{V}}{U}\right\}.$$

The smallest c attaining nominal coverage at level $1-\alpha$ over a set of replicates is therefore the $(1-\alpha)$ quantile of that ratio across them, and its Monte Carlo interval comes from the same order statistics. No search over candidate factors is needed, and the estimate inherits the replicate count directly: the plane carries 20 cells at 500 replicates each, over K/n from 0.0025 to 0.500, on two data-generating worlds rather than one (Table S34).

S9.2. A monotone fit, not a parametric one

The dependence of c on K/n is well summarised by a log-linear fit whose slope is 0.122, with a standard error of 0.006: the widening a pair needs grows with its register's cardinality relative to its training set, as the mechanism of Section 4.1 predicts. That fit is quoted because it is the interpretable summary, but it is *not* what the corpus is calibrated with. The parametric fit sits below the measured factor at the smallest ratios, and using it would under-correct exactly the pairs whose bands are narrowest — the pairs where an anti-conservative band does the most damage. The corpus is therefore calibrated with an isotonic regression through the measured factors, which is monotone by construction and interpolates the measurements rather than smoothing through them. Table S34 prints the two curves side by side.

S9.3. What the calibration does and does not correct

The factor is estimated on a *pointwise* interval in a simulation and applied to a *simultaneous* band on real logs. It corrects the scale of the studentised statistic, which is the channel the plane identifies. It does not correct a change in the shape of the maximum's distribution, and it assumes that the simulated worlds' dependence on K/n transfers to logs whose dependence structure and signal are not the simulation's. Because that transfer is an assumption, Section 6.4 reports the corpus counts at three settings — nominal, calibrated, and every factor pushed to the upper end of its own Monte Carlo interval — and Section 11 carries the transfer as a limitation.

S10. Results the article summarises

S10.1. The pipeline axis is two axes

Table 4's "learner" column is not a learner axis. Its levels are pipelines: a model family together with an encoding and, in one case, a calibration. On 15 of the 19 pairs only two of them run, so the index is a statement about a two-level axis whose levels differ in two things at once.

On the case study's log the budget allows the two to be crossed. Both model families — penalised logistic regression and histogram gradient boosting — are run against all three encodings: one-hot, frequency, and cross-fitted target encoding, on both of the log's targets. That is 12 complete factorial pipelines, and the decomposition of Section 4.5 then carries a family axis and an encoding axis separately.

Three things follow, and the third is the reason the confound mattered.

The encoding is the larger of the two. It carries 10.1% of the variance in the increment against the family's 5.2%. How the register is turned into columns matters about twice as much as which model consumes them, and it is the choice more often left unstated.

Their interaction is as large as either main effect. The family–encoding two-way component carries 9.7%, the largest two-way term in the crossed decomposition. A gradient-boosting model and a logistic model do not respond to a change of encoding in the same way, which is precisely what an axis that fuses them cannot express.

So the fused index is not the sum of two indices. It is a statement about a two-level axis whose levels move two things at once, and Table 4's pipeline column should be read as that and nothing more. The crossing has not been re-run on the other 15 pairs, and Section 11 carries that as a limitation rather than assuming the case study's split transfers.

S10.2. Decision rules, in sample and out

Table S20 compares the four rules of Section 4.7 — *one-number*, *uniform*, *majority* and *weighted mean* — under the loss defined there, inside one in-

log	target	encoding	family	quality level	rung	split	higher-order
BPIC14	duration	0.091	0.066	0.065	0.492	0.014	0.271
BPIC14	handover	0.125	0.049	0.109	0.430	0.009	0.276

Table S4: The decomposition with the model family and the encoding as separate axes, on the case study's log where all six pipelines run. First-order shares, median over the five instruments, under the equal-level measure, then the total higher-order share. The 'learner' axis of Table 4 is the two of them together.

strument at a time and in that instrument's own units. No average in this paper crosses an instrument.

The weighted-mean rule is optimal in sample by construction — its excess regret is exactly zero everywhere, as the lemma requires — so the comparison that matters is out of sample. Choosing the action on a random half of the admissible cells and paying the loss on the other half, the one-number rule costs 0.00450 AUC of expected excess regret against 0.00011 for the majority rule and 0.00002 for the weighted mean. The part of the ordering that holds everywhere is that the one-number rule is worse than the majority rule, in 5 instruments of 5; the full three-way ordering holds in 4 of them.

Three results run against the surface's advocate and are reported as such. The conservative rule — adopt only where the whole surface is positive — is the worst of the four in 4 of 5 instruments, because it declines so often that the performance it forgoes exceeds the performance the one-number rule destroys; the exception is Nagelkerke R^2. The median excess regret of the one-number rule is *zero* in every instrument: on most pairs every rule picks the same action. And under rolling-origin folds — choosing the action on the earlier folds and paying on the later ones, a harder test because the surface itself moves in time — the ordering does not hold: one number costs 0.00885 against the majority rule's 0.01922. Leaving one log out and asking whether a pairwise ordering established on the rest survives on the held-out log, it does on 57% of the 79 comparisons in which the two rules differ at all, which is near chance.

What we conclude from this, and what we do not. We do not conclude that a surface report is a better decision rule than a number. On this corpus the decision-optimal collapse is the weighted mean, that is an algebraic identity in sample and a small and unstable advantage out of it, and on the one utility with an operational meaning the rules mostly tie (Section 8.4). The case for reporting the surface rests on the decomposition and on the sign disagreement, not on the regret comparison, and the regret comparison is reported here as a check on the former rather than as a result in its own right.

S10.3. Calibration, unseen categories and rolling origin

Table S21 reports, for each pair at the reference cell, the calibration intercept and slope, the Brier score, and the share of test cases carrying at least one level unseen in training. The unseen-category rate is the mechanism behind much of the register's behaviour: a register whose identities do not recur across the split cannot help a model that has never seen them, and the rate ranges from 0% to 87%. Calibration is reported because discrimination alone is not a sufficient description of a model [2]; slopes below one indicate the overfitting that high-cardinality one-hot encodings produce, and are the reason a calibrated learner is one of the four.

The frequency-encoded pipeline is the one that is visibly miscalibrated, with a slope well below one, and it is also the one whose increment differs most from the others on the two proper scoring rules. That is the interaction between the pipeline axis and the metric axis, drawn.

Rolling-origin results are in `results/s01_surface.csv` under `split = rolling`k and enter the decomposition as the split axis; Supplement S5 prints them per fold. The single holdout is not the best estimate of the increment, and reporting it alone would be a limitation.

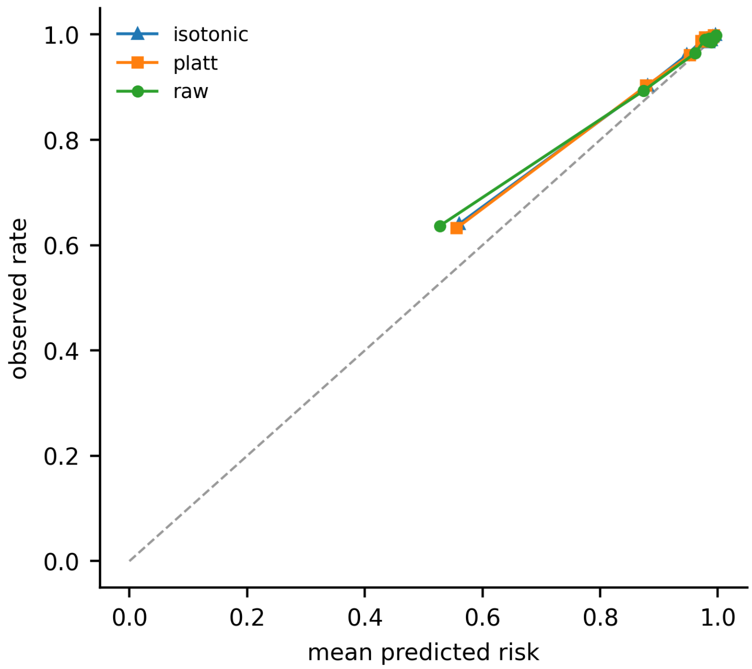

Figure S1: Calibration at the reference cell on the case study's log, by decile of predicted risk. The two proper scoring rules on the metric axis — Brier skill and the log-loss skill — read this picture; the two rank-based ones cannot see it at all (Proposition 2). A pipeline whose discrimination is adequate and whose calibration is not therefore moves the increment on some instruments and not others, which is one concrete mechanism behind the metric axis's contribution to Table 4.

S11. Tables the article summarises

The article prints the seven tables a reader needs to follow its argument. This section carries the ones it summarises: the exclusion ledger, the construct audit, the surface denominator, the planned contrasts, the region labels, the ITSM family split, the sign-flip attribution, the cohort decomposition, the axis-kind partition, the specification-curve comparison, the desk utility, the tipping curve, the decision-curve corpus, the declared measures, the four decision rules, the calibration diagnostics, the availability evidence, the severity sweeps, the absorption ladders, and the simulation in full. Every one is generated by `scripts/make_numbers.py` from the result files, and none is typed by hand.

log	domain	code	detail
Helpdesk	itsm	NO_HEADROOM	handover/primary prev=0.9993
Helpdesk	itsm	NO_HEADROOM	handover/g=workgroup prev=0.0011
BPIC13_open	itsm	SMALL_N	n=819
BPIC13_closed	itsm	NO_TEST_VARIATION	duration
BPIC12	lending	NO_G	n=13087
BPIC17	lending	NO_HEADROOM	handover/primary prev=0.9977
BPIC19	procurement	NO_HEADROOM	handover/primary prev=0.9631
BPIC15_2	permitting	SMALL_N	n=832
BPIC15_5	permitting	NO_HEADROOM	handover/primary prev=0.9593
BPIC20_domestic	expenses	NO_G	n=10500
BPIC20_intl	expenses	NO_G	n=6449
BPIC20_prepaid	expenses	NO_G	n=2099
BPIC20_permit	expenses	NO_G	n=7065
BPIC20_rfp	expenses	NO_G	n=6886
Sepsis	healthcare	NO_HEADROOM	handover/primary prev=1.0000
HospitalBilling	healthcare	NO_F	n=100000
RoadFines	enforcement	NO_HEADROOM	handover/primary prev=0.0037
BPIC11	healthcare	HELD_OUT	downloaded as a held-out log for the out-of-corpus test; never offered to the admission rules
BPIC18	agriculture	NO_TEST_VARIATION	held-out set: duration
BPIC18	agriculture	NO_HEADROOM	held-out set: handover prev=1.0000

Table S5: Every downloaded log that is not among the 19 admitted log–target pairs, with its registered exclusion code. HELD_OUT names the logs downloaded as a held-out set for a separate test and never offered to the admission rules; every other code is a rule of Section 5. Some logs appear here under one target and among the admitted pairs under the other, which is what a per-pair admission rule does.

S11.1. The corpus

S11.2. Regions, sign flips and the cohort decomposition

S11.3. Measures, rules and calibration

S11.4. The case study

S11.5. The same objects on the registered cohort

Section 7.1 shows that this log gives two answers, and that the difference is the target rather than the cohort. Section 7 prints the estate's own cohort and reassignment target throughout. These are the same objects on the registered cohort and the handover target, so that a reader can see both without either being the one the article quotes.

S11.6. The simulation in full

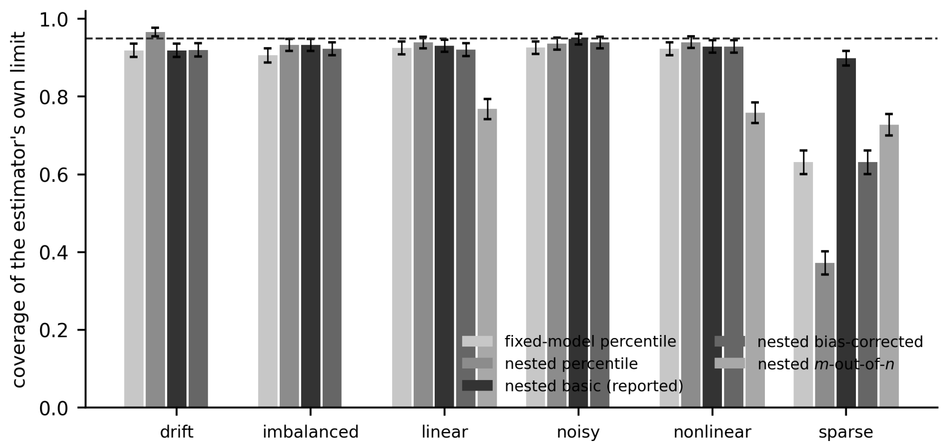

Figure S2: Coverage of the estimator's own limit in six simulated worlds, by interval construction, with Monte Carlo error bars. The dashed line is nominal. The fixed-model percentile interval is the conventional one; the nested basic interval is the one this paper reports. The sparse world is where they disagree by more than a few points, and the nested percentile bar there is the failure the basic construction repairs.

log	domain	target	register	free field	why it behaves as a register	threat to construct validity
RoadFines	enforcement	duration	article	org:resource	article of law, reused across fines	semantics legal, not operational
Sepsis	healthcare	duration	Diagnose	org:group	diagnosis vocabulary: reused, not maintained	no maintenance process; quality axis hypothetical
BPIC13_closed	itsm	handover	product	org:resource	maintained CI or customer register	none beyond domain: closest to the case study
BPIC13_incidents	itsm	duration	product	org:resource	maintained CI or customer register	none beyond domain: closest to the case study
BPIC13_incidents	itsm	handover	product	org:resource	maintained CI or customer register	none beyond domain: closest to the case study
BPIC14	itsm	duration	CI Name (aff)	assignment group	maintained CI or customer register	none beyond domain: closest to the case study
BPIC14	itsm	handover	CI Name (aff)	assignment group	maintained CI or customer register	none beyond domain: closest to the case study
Helpdesk	itsm	duration	customer	Resource	maintained CI or customer register	none beyond domain: closest to the case study
UCI498	itsm	duration	caller_id	assigned_to	maintained CI or customer register	none beyond domain: closest to the case study
UCI498	itsm	handover	caller_id	assigned_to	maintained CI or customer register	none beyond domain: closest to the case study
BPIC17	lending	duration	case:RequestedAmount	org:resource	application or offer reference	near-unique per case, so reuse is thin
BPIC15_1	permitting	duration	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_1	permitting	handover	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_3	permitting	duration	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_3	permitting	handover	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_4	permitting	duration	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_4	permitting	handover	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_5	permitting	duration	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC19	procurement	duration	case:Vendor	org:resource	item or vendor catalogue entry	target is a duration, not a routing decision

Table S6: Construct validity, per log–target pair: the field playing the register role, why it behaves as a maintained and reused entity, the free field, and the specific threat to construct validity the pair carries. These are not the same construct, which is why the corpus is a benchmark of specification sensitivity rather than a replication.

log	target	learners	splits	quality	rungs	metrics	declared scalar	computed scalar	model fits	duplicates	missing	inference cells	draws	register levels K	K / n train
BPIC13_closed	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	337	0.324
BPIC13_incidents	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	704	0.133
BPIC13_incidents	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	704	0.133
BPIC14	duration	4	6	10	4	5	4800	4800	1200	0	0	96	150	3019	0.093
BPIC14	handover	4	6	10	4	5	4800	4800	1200	0	0	96	150	3019	0.093
BPIC15_1	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	89	0.106
BPIC15_1	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	89	0.106
BPIC15_3	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	94	0.095
BPIC15_3	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	94	0.095
BPIC15_4	duration	2	6	6	3	5	1080	1080	288	0	0	36	150	53	0.072
BPIC15_4	handover	2	6	6	3	5	1080	1080	288	0	0	36	150	53	0.072
BPIC15_5	duration	2	6	6	3	5	1080	1080	288	0	0	36	150	107	0.132
BPIC17	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	701	0.032
BPIC19	duration	3	6	6	3	5	1620	1620	432	0	0	24	40	1975	0.011
Helpdesk	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	394	0.123
RoadFines	duration	3	6	6	3	5	1620	1620	432	0	0	24	40	66	0.001
Sepsis	duration	2	6	6	3	5	1080	1080	288	0	0	36	150	147	0.200
UCI498	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	5245	0.301
UCI498	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	5245	0.301

Table S7: The surface denominator audit. For every log–target pair: the declared level count on each axis, then two counts *on the same basis* — the admissible scalar cells the declaration multiplies out, and the same cells counted in the surface file — which are equal on every pair. 'model fits' is a different quantity and is printed separately: it counts fitted arms before the expansion over instruments and *includes* the intercept-only rung, which the master surface carries so that the surface is complete and which no admissible set contains. Then duplicates, missing combinations, the size and draw count of the inference surface, the register's cardinality and its ratio to the training row count, which Section 10.3 shows is what governs whether the bands cover at their nominal level.

contrast	cell a	cell b	estimate	interval	p Holm	Holm	resolved
PC1	intake	–	0.25718	[+0.23556, +0.27809]	0.00250	True	True
PC2	intake + g	–	0.16727	[+0.14456, +0.20735]	0.00250	True	True
PC3	intake + g + km	–	0.00768	[-0.00059, +0.01210]	0.00250	True	False
PC4	intake	intake + g	0.08991	[+0.04844, +0.11706]	0.00250	True	True
PC5	intake + g	intake + g, long tail 50%	0.06339	[+0.05368, +0.07740]	0.00250	True	True

Table S8: The five final-round *planned* contrasts (PC1–PC5), stated in words in Section 4.4. Every one is on the REGISTERED cohort (46,606 cases) and the handover target — not the case study's cohort and reassignment target, which Section 7.1 reconciles against these — at incident creation, with the register clean, under one-hot logistic regression on the single temporal holdout in ROC AUC; PC3 additionally admits the knowledge reference and PC5's second cell carries the quality named in it. p-values are ONE-SIDED bootstrap values from the plus-one estimator $(k+1)/(B+1)$ with a Holm step-down correction; the intervals are TWO-SIDED basic intervals from the same draws, and the last column says whether the interval resolves — which is the criterion this paper uses and is not the same as the p-value's. PC3 is the contrast on which they differ, and Section 4.4 derives why. The contrasts are not confirmatory: they were fixed before this analysis round, not before the data were seen.

log	target	cells	beneficial	harmful	unresolved	rho	region	q	label under the narrow family
BPIC13_closed	handover	180	0	1	179	-0.006	conditionally harmful	3.338	sign-changing
BPIC13_incidents	duration	180	21	0	159	0.117	conditionally beneficial	3.354	sign-changing
BPIC13_incidents	handover	180	97	0	83	0.539	conditionally beneficial	3.330	conditionally beneficial
BPIC14	duration	480	337	0	143	0.702	conditionally beneficial	3.492	conditionally beneficial
BPIC14	handover	480	279	2	199	0.577	sign-changing	3.569	sign-changing
BPIC15_1	duration	180	6	0	174	0.033	conditionally beneficial	3.339	conditionally beneficial
BPIC15_1	handover	180	2	0	178	0.011	conditionally beneficial	3.358	sign-changing
BPIC15_3	duration	180	40	0	140	0.222	conditionally beneficial	3.335	sign-changing
BPIC15_3	handover	180	2	0	178	0.011	conditionally beneficial	3.318	sign-changing
BPIC15_4	duration	180	0	0	180	0.000	unresolved	3.380	sign-changing
BPIC15_4	handover	180	0	0	180	0.000	unresolved	3.425	sign-changing
BPIC15_5	duration	180	4	3	173	0.006	sign-changing	3.395	sign-changing
BPIC17	duration	180	35	0	145	0.194	conditionally beneficial	3.252	conditionally beneficial
BPIC19	duration	120	14	28	78	-0.117	sign-changing	3.317	sign-changing
Helpdesk	duration	180	0	8	172	-0.044	conditionally harmful	3.312	sign-changing
RoadFines	duration	120	47	20	53	0.225	sign-changing	3.140	sign-changing
Sepsis	duration	180	106	0	74	0.589	conditionally beneficial	3.320	conditionally beneficial
UCI498	duration	180	58	0	122	0.322	conditionally beneficial	3.257	conditionally beneficial
UCI498	handover	180	40	0	140	0.222	conditionally beneficial	3.329	sign-changing

Table S9: Resolution regions from the WHOLE-SURFACE simultaneous band at its NOMINAL critical value, with that value q and, for comparison, the label a narrower within-instrument family would have given the same draws. **These are not the labels the article quotes**: the article's regions and ρ are computed under the coverage-calibrated critical value and are in Table 5. This table is printed so that the cost of the wider FAMILY is visible separately from the cost of the CALIBRATION.

statistic	all 19	itsm 8	other 11
baseline spread (AUC)	0.073 [0.024, 0.097]	0.070 [0.014, 0.132]	0.088 [0.022, 0.097]
higher-order share	0.560 [0.409, 0.608]	0.365 [0.233, 0.450]	0.608 [0.532, 0.732]
largest first-order index	0.287 [0.185, 0.352]	0.355 [0.278, 0.358]	0.201 [0.151, 0.305]
misreport, all cells	0.328 [0.243, 0.452]	0.271 [0.244, 0.481]	0.353 [0.191, 0.617]
misreport, analyst latitude only	0.200 [0.100, 0.467]	0.194 [0.067, 0.367]	0.400 [0.067, 0.600]
misreport, magnitude-weighted	0.502 [0.102, 0.623]	0.296 [0.025, 0.623]	0.507 [0.139, 0.640]
misreport, resolved cells (pooled)	0.075	0.036	0.222
robustness index rho (calibrated)	0.017 [0.000, 0.182]	0.069 [-0.006, 0.552]	0.000 [0.000, 0.182]
one-number excess regret (AUC)	0.000 [0.000, 0.003]	0.000 [0.000, 0.011]	0.000 [0.000, 0.005]

Table S10: Every headline corpus statistic on three populations: all nineteen log–target pairs, the eight from IT service management where the register is a maintained configuration or customer register, and the eleven where it is not. Each cell is the statistic with a 95% percentile interval from a cluster bootstrap that resamples LOGS, because two targets on the same log share a cohort and are not two independent observations. The resolved-cell row is a pooled ratio — disagreeing cells over resolved cells, summed across the population — and not a median of per-pair rates, which on a population where most pairs resolve little prints zero and says nothing.

log	target	learner	metric	quality	rung	split
BPIC13_closed	handover	0.287	0.214	0.234	0.222	0.401
BPIC13_incidents	duration	0.374	0.249	0.204	0.230	0.405
BPIC13_incidents	handover	0.174	0.089	0.111	0.100	0.181
BPIC14	duration	0.155	0.051	0.121	0.150	0.089
BPIC14	handover	0.426	0.096	0.115	0.139	0.113
BPIC15_1	duration	0.383	0.217	0.228	0.348	0.446
BPIC15_1	handover	0.313	0.168	0.232	0.263	0.331
BPIC15_3	duration	0.337	0.144	0.164	0.187	0.250
BPIC15_3	handover	0.365	0.302	0.217	0.276	0.449
BPIC15_4	duration	0.431	0.169	0.238	0.196	0.501
BPIC15_4	handover	0.450	0.254	0.255	0.287	0.454
BPIC15_5	duration	0.426	0.194	0.221	0.309	0.335
BPIC17	duration	0.291	0.219	0.257	0.224	0.397
BPIC19	duration	0.183	0.217	0.160	0.079	0.451
Helpdesk	duration	0.252	0.273	0.181	0.131	0.284
RoadFines	duration	0.120	0.172	0.181	0.112	0.221
Sepsis	duration	0.113	0.066	0.058	0.063	0.089
UCI498	duration	0.300	0.197	0.199	0.383	0.384
UCI498	handover	0.333	0.206	0.230	0.489	0.289

Table S11: Sign flips attributable to each axis: the share of pairs of cells that agree on every other axis and disagree on the sign of the increment.

effect	B intake	B intake g	B intake g km
cohort	0.000	0.000	0.000
cohort x target	0.000	0.000	0.007
target	1.000	1.000	0.993

Table S12: The exact two-factor decomposition of the discrepancy. Share of the variance of V over the cohort $\times$ target design, at each rung of the ladder, at a declared tie-break. With two levels per factor the decomposition is exact.

log	target	lat. raw	res. raw	ctf. raw	h.o. raw	lat. hdrm	res. hdrm	ctf. hdrm	h.o. hdrm
BPIC13_closed	handover	0.223	0.203	0.029	0.546	0.192	0.111	0.014	0.683
BPIC13_incidents	duration	0.111	0.384	0.029	0.475	0.101	0.158	0.011	0.729
BPIC13_incidents	handover	0.174	0.172	0.099	0.555	0.080	0.094	0.031	0.795
BPIC14	duration	0.606	0.013	0.151	0.231	0.641	0.008	0.104	0.247
BPIC14	handover	0.631	0.010	0.127	0.232	0.612	0.007	0.065	0.316
BPIC15_1	duration	0.084	0.263	0.039	0.613	0.160	0.141	0.005	0.694
BPIC15_1	handover	0.106	0.079	0.036	0.779	0.043	0.060	0.027	0.869
BPIC15_3	duration	0.037	0.287	0.081	0.595	0.107	0.184	0.068	0.641
BPIC15_3	handover	0.128	0.189	0.004	0.679	0.088	0.125	0.001	0.786
BPIC15_4	duration	0.083	0.111	0.025	0.781	0.062	0.156	0.007	0.776
BPIC15_4	handover	0.090	0.050	0.031	0.829	0.089	0.017	0.038	0.856
BPIC15_5	duration	0.157	0.036	0.042	0.766	0.231	0.043	0.020	0.705
BPIC17	duration	0.001	0.346	0.062	0.591	0.408	0.087	0.046	0.459
BPIC19	duration	0.017	0.491	0.011	0.481	0.165	0.374	0.010	0.451
Helpdesk	duration	0.287	0.222	0.086	0.405	0.356	0.151	0.065	0.428
RoadFines	duration	0.065	0.192	0.289	0.454	0.534	0.016	0.090	0.360
Sepsis	duration	0.127	0.134	0.145	0.594	0.366	0.061	0.075	0.498
UCI498	duration	0.107	0.399	0.030	0.465	0.343	0.116	0.008	0.533
UCI498	handover	0.396	0.136	0.071	0.398	0.427	0.057	0.025	0.491

Table S13: The variance decomposition partitioned by what KIND of choice each axis is, under both declared scales. *Analyst latitude* is the pipeline and the baseline rung — and, on the second scale only, the instrument — things an analyst chooses and could have chosen otherwise. *Resampling* is the split, five of whose six levels are expanding-origin folds on the same data and are therefore different samples rather than different analyses. *Counterfactual* is the register-quality condition, which is a different state of the world. The remainder belongs to no single axis. *Columns.* lat. is analyst latitude, res. resampling, ctf. the counterfactual and h.o. the higher-order remainder; raw and hdrm are the two scales. *Which scale, and why there are two.* raw, within instrument is the paper's primary scale: the decomposition is taken separately inside each of the five instruments, on that instrument's own units, and the median is reported — so the instrument is a stratum here and its own contribution appears in no column. headroom, instrument as an axis is the secondary scale of Section 4.5: raw AUC and raw average precision do not share units, so making the instrument a sixth axis requires the headroom rescaling. Every row is the equal-level measure. The two scales order *analyst latitude* against *resampling* oppositely, which is a property of the scale and not of the corpus; Section 6.6 says so rather than choosing one.

axis varied	reference level	pairs	median rate	min over pairs	max over pairs
the declared reference	–	19	0.328	0.056	0.623
pipeline	boosting, target-encoded	19	0.353	0.056	0.716
pipeline	boosting, isotonic	19	0.328	0.056	0.640
pipeline	logistic, frequency-encoded	19	0.353	0.056	0.756
instrument	average precision	19	0.328	0.056	0.623
instrument	Brier skill	19	0.382	0.056	0.672
instrument	log-loss skill	19	0.382	0.056	0.672
instrument	nagelkerke	19	0.382	0.056	0.672
baseline rung	half the intake block	19	0.328	0.056	0.623
baseline rung	intake only	19	0.328	0.056	0.623
baseline rung	intake, group and knowledge	19	0.353	0.056	0.894
split	rolling1	19	0.353	0.056	0.743
split	rolling2	19	0.328	0.056	0.493
split	rolling3	19	0.353	0.056	0.743
split	rolling4	19	0.328	0.056	0.623
split	rolling5	19	0.384	0.056	0.716
register quality	clean at 1.0	19	0.328	0.056	0.623
register quality	corrupt at 0.05	19	0.328	0.056	0.623
register quality	corrupt at 0.15	19	0.328	0.056	0.647
register quality	duplicate at 0.15	19	0.328	0.056	0.623
register quality	mask common at 0.5	19	0.328	0.056	0.623
register quality	mask random at 0.5	19	0.328	0.056	0.623
register quality	mask rare at 0.25	19	0.382	0.056	0.889
register quality	mask rare at 0.5	19	0.328	0.056	0.623
register quality	mask rare at 0.75	19	0.328	0.056	0.623
register quality	stale at 1.0	19	0.328	0.056	0.623

Table S14: The headline sign-disagreement rate under a different declared reference specification. Each row moves the reference ONE axis at a time and holds the rest at the declared cell, then recomputes the all-cells rate on every pair under the equal-level measure; the first row is the paper's own reference. The reference supplies the sign the other cells are counted as agreeing or disagreeing with, so moving it moves which cells disagree and nothing else. The median column is the corpus median the article quotes; the two beside it are over pairs and show that the spread within the corpus is far larger than the spread across reference specifications.

log	target	cases	capped	observed median	null median	p (median)	share positive	SCA verdict	region	rho
BPIC13_closed	handover	1487	False	-0.010	-0.008	0.327	0.208	not distinguishable from null	unresolved	0.000
BPIC13_incidents	duration	7554	False	0.010	-0.012	0.673	0.625	not distinguishable from null	conditionally beneficial	0.017
BPIC13_incidents	handover	7554	False	0.031	-0.006	0.010	0.833	non-null at 5%	conditionally beneficial	0.400
BPIC14	duration	20000	True	0.091	-0.001	0.010	1.000	non-null at 5%	conditionally beneficial	0.692
BPIC14	handover	20000	True	0.035	0.000	0.010	0.750	non-null at 5%	sign-changing	0.552
BPIC15_1	duration	1199	False	0.030	0.001	0.010	0.792	non-null at 5%	conditionally beneficial	0.011
BPIC15_1	handover	1199	False	0.002	-0.002	0.653	0.708	not distinguishable from null	unresolved	0.000
BPIC15_3	duration	1409	False	0.119	0.003	0.010	0.958	non-null at 5%	conditionally beneficial	0.056
BPIC15_3	handover	1409	False	0.023	-0.005	0.020	0.625	non-null at 5%	unresolved	0.000
BPIC15_4	duration	1053	False	-0.030	-0.004	0.010	0.333	non-null at 5%	unresolved	0.000
BPIC15_4	handover	1053	False	-0.018	-0.011	0.158	0.250	not distinguishable from null	unresolved	0.000
BPIC15_5	duration	1156	False	0.043	-0.001	0.010	0.833	non-null at 5%	unresolved	0.000
BPIC17	duration	20000	True	0.006	-0.001	0.030	0.708	non-null at 5%	conditionally beneficial	0.139
BPIC19	duration	20000	True	0.051	-0.001	0.010	1.000	non-null at 5%	sign-changing	-0.083
Helpdesk	duration	4580	False	-0.013	-0.008	0.109	0.208	not distinguishable from null	conditionally harmful	-0.011
RoadFines	duration	20000	True	0.016	0.000	0.010	0.792	non-null at 5%	sign-changing	0.225
Sepsis	duration	1050	False	0.154	-0.003	0.010	1.000	non-null at 5%	conditionally beneficial	0.283
UCI498	duration	20000	True	0.005	-0.005	0.396	0.625	not distinguishable from null	conditionally beneficial	0.094
UCI498	handover	20000	True	0.000	-0.002	1.000	0.500	not distinguishable from null	conditionally beneficial	0.044

Table S15: Specification-curve analysis as practised, beside this paper's region label. The permutation test asks whether the surface as a whole differs from one with no signal in it; the region asks, of each cell, whether the sign of that cell's own increment is determined. Where the two part company is what the extra machinery is for. Both verdicts are computed on the declared 24-cell sub-surface at 100 permutations; 'cases' is the number of cases the permutation test ran on, which is the pair's own count unless the declared cap of 20,000 bound on it. The region column is the coverage-calibrated label of Section 6.4, computed on the full data.

threshold measure	majority	one-number	uniform-beneficial	weighted-mean	pairs where the rules differ
desk distribution	0	0	0.0477	0	11
fixed 1:19	0	0	0.0000	0	5
fixed 1:2	0	0	0.3251	0	10
fixed 1:4	0	0	0.2127	0	11
fixed 1:9	0	0	0.0000	0	7
uniform over the declared range	0	0	1.3468	0	13

Table S16: Median excess regret of each decision rule, in net benefit per thousand cases, on the models that pass the registered calibration rule. The desk distribution places its mass on the exchange rates a service desk plausibly holds; the fixed rows report single exchange rates so that a reader who rejects the distribution still has a number. The last column counts the pairs on which the four rules do not all lose the same.

lambda posthoc	V auc	sd	n seeds
0.000	-0.003	0.000	5
0.100	0.000	0.001	5
0.200	0.006	0.002	5
0.300	0.014	0.003	5
0.400	0.024	0.003	5
0.500	0.033	0.003	5
0.600	0.046	0.003	5
0.700	0.059	0.004	5
0.800	0.076	0.003	5
0.900	0.091	0.002	5
1.000	0.104	0.000	5

Table S17: The register's increment against the share of knowledge references assumed post-hoc.

metric	V	se	sim lo	sim hi	sim resolved
nb_0.050	0.000	0.000	0.000	0.000	False
nb_0.075	0.000	0.000	0.000	0.000	False
nb_0.100	0.000	0.000	0.000	0.000	False
nb_0.125	0.000	0.000	-0.001	0.001	False
nb_0.150	0.001	0.001	-0.002	0.004	False
nb_0.175	0.003	0.001	0.001	0.008	True
nb_0.200	0.003	0.001	0.000	0.007	False
nb_0.225	0.004	0.001	0.001	0.007	True
nb_0.250	0.004	0.001	0.001	0.008	True
nb_0.275	0.005	0.001	0.001	0.008	True
nb_0.300	0.005	0.001	0.001	0.009	True
nb_0.325	0.006	0.001	0.002	0.010	True
nb_0.350	0.007	0.001	0.002	0.012	True
nb_0.375	0.008	0.001	0.002	0.013	True
nb_0.400	0.010	0.001	0.004	0.016	True
nb_0.425	0.011	0.002	0.006	0.019	True
nb_0.450	0.012	0.002	0.006	0.020	True
nb_0.475	0.014	0.002	0.007	0.023	True
nb_0.500	0.016	0.005	-0.003	0.038	False
nb_0.525	0.004	0.007	-0.023	0.031	False
nb_0.550	0.007	0.003	-0.004	0.019	False
nb_0.575	0.010	0.012	-0.041	0.058	False
nb_0.600	0.037	0.006	0.016	0.060	True
nb_0.625	0.033	0.005	0.011	0.055	True
nb_0.650	0.036	0.005	0.020	0.064	True
nb_0.675	0.030	0.005	0.010	0.051	True
nb_0.700	0.034	0.004	0.020	0.052	True
nb_0.725	0.037	0.004	0.024	0.056	True
nb_0.750	0.037	0.004	0.020	0.054	True
nb_0.775	0.043	0.005	0.025	0.064	True
nb_0.800	0.050	0.006	0.030	0.077	True

Table S18: The decision curve of Figure 5 as a table: BPIC14, the registered handover target, one-hot logistic regression, the clean register, the single temporal holdout, the intake block plus the free field as baseline. Every row is one operating point on RAW scores, and is therefore a score-threshold sensitivity analysis in the sense of Section 8.2 rather than a decision curve. The bands are the whole-family simultaneous max-t construction over this pair's declared decision-curve family, and 'sim resolved' says whether that band excludes zero. This family is NOT coverage-calibrated; Section 8.3 gives the counts under the empirical critical value beside these.

measure	description	median	min	max	largest first order median	n pairs first order exceeds	n pairs
equal-level	every level of every axis equally likely	0.560	0.233	0.793	0.287	6.000	19.000
reference	half the mass on the reference level of each axis	0.496	0.157	0.709	0.299	6.000	19.000
concentrated	eighty per cent on the reference level	0.317	0.065	0.638	0.392	12.000	19.000
envelope	200 Dirichlet draws of the level weights, primary pair	0.181	0.098	0.347	–	–	–

Table S19: The declared measures over specifications, and the total higher-order variance share under each. The magnitude moves with the measure; the qualitative conclusion, that no first-order index approaches the higher-order remainder, does not.

metric	unit	majority	one-number	uniform-beneficial	weighted-mean	majority (n suboptimal)	one-number (n suboptimal)	uniform-beneficial (n suboptimal)	weighted-mean (n suboptimal)
ap	average precision	0.000	0.002	0.027	0	1	3	17	0
auc	AUC	0.000	0.005	0.026	0	0	5	14	0
brier_skill	Brier skill	0.000	0.002	0.031	0	3	4	9	0
logloss_skill	log-loss skill	0.003	0.004	0.023	0	3	3	8	0
nagelkerke	Nagelkerke R2	0.025	0.042	0.037	0	6	6	7	0

Table S20: Four decision rules, INSIDE one instrument at a time. Mean excess regret across log–target pairs in that instrument's own units, and the number of pairs on which each rule is strictly suboptimal. No column averages across instruments, because there is no unit in which such an average would be expressed.

log	target	n train	n test	prev test	cal intercept	cal slope	brier	unseen
BPIC13_closed	handover	1040	447	0.275	-0.368	0.916	0.170	0.407
BPIC13_incidents	duration	5287	2267	0.154	-1.783	0.571	0.216	0.105
BPIC13_incidents	handover	5287	2267	0.616	-0.925	0.943	0.167	0.105
BPIC14	duration	32624	13982	0.427	-0.259	0.938	0.202	0.053
BPIC14	handover	32624	13982	0.945	0.378	0.994	0.037	0.053
BPIC15_1	duration	839	360	0.442	-1.204	0.997	0.228	0.753
BPIC15_1	handover	839	360	0.839	-0.061	1.255	0.066	0.753
BPIC15_3	duration	986	423	0.456	-0.746	0.928	0.214	0.095
BPIC15_3	handover	986	423	0.905	-0.649	1.437	0.069	0.095
BPIC15_4	duration	737	316	0.225	-1.927	0.029	0.351	0.649
BPIC15_4	handover	737	316	0.867	-1.349	0.302	0.119	0.649
BPIC15_5	duration	809	347	0.285	-0.793	0.290	0.244	0.867
BPIC17	duration	22056	9453	0.494	-0.060	0.758	0.244	0.060
BPIC19	duration	176213	75521	0.162	-3.163	0.278	0.352	0.299
Helpdesk	duration	3206	1374	0.451	-0.149	0.475	0.245	0.229
RoadFines	duration	105259	45111	0.429	-0.312	0.626	0.244	0.368
Sepsis	duration	735	315	0.444	-0.307	0.716	0.192	0.079
UCI498	duration	17442	7476	0.286	-0.619	0.811	0.176	0.192
UCI498	handover	17442	7476	0.420	0.017	1.230	0.071	0.192

Table S21: Calibration and unseen-category rates at the reference cell, one row per log–target pair in the same order as every other per-pair table. Every row is on the cohort and target the registered rules of Section 5 define, BPIC14's included; the case study's own cohort is smaller and its counts are in Section 7.1.

log	target	raw slope	platt slope	isotonic slope	raw ece	platt ece	isotonic ece
BPIC13_closed	handover	0.332	1.005	0.886	0.149	0.120	0.096
BPIC13_incidents	duration	0.575	1.052	0.902	0.295	0.304	0.296
BPIC13_incidents	handover	0.743	1.334	1.049	0.140	0.149	0.142
BPIC14	duration	0.945	1.117	1.050	0.056	0.056	0.056
BPIC14	handover	0.952	1.049	0.995	0.017	0.015	0.015
BPIC15_1	duration	0.731	1.812	1.118	0.208	0.172	0.152
BPIC15_1	handover	0.778	1.654	1.466	0.075	0.066	0.050
BPIC15_3	duration	0.886	1.225	1.109	0.190	0.152	0.152
BPIC15_3	handover	0.450	1.247	0.904	0.071	0.057	0.039
BPIC15_4	duration	0.025	0.028	-0.013	0.364	0.313	0.340
BPIC15_4	handover	0.282	0.701	0.456	0.086	0.071	0.080
BPIC15_5	duration	0.263	0.853	0.414	0.252	0.209	0.211
BPIC17	duration	0.868	1.165	1.031	0.023	0.019	0.017
BPIC19	duration	0.293	0.355	0.372	0.339	0.338	0.338
Helpdesk	duration	0.490	0.731	0.622	0.088	0.066	0.065
RoadFines	duration	0.688	0.720	1.377	0.081	0.086	0.081
Sepsis	duration	0.456	0.830	0.554	0.147	0.086	0.089
UCI498	duration	0.882	1.534	1.333	0.104	0.185	0.175
UCI498	handover	1.077	1.453	1.203	0.041	0.063	0.047

Table S22: Calibration before and after a training-only recalibration, one row per log–target pair: the median calibration slope and expected calibration error over the arms of the baseline ladder, raw and under each calibrator. A slope far from one is a score whose thresholds are not cost ratios; only curves computed on calibrated probabilities carry a probability-threshold reading.

evidence	value	establishes
interactions in the file	147004.000	the population the knowledge reference is written in
interactions that never became incidents	94250.000	the field is written by the service-desk process, not by the incident process
knowledge reference populated on those	1.000	the same, quantitatively
interactions CLOSED before the incident was opened	5.000	airtight pre-incident availability, on five cases, which is not enough to rest a headline on
interactions whose recorded handling time had elapsed before the incident was opened	41413.000	that the desk had worked the interaction, which is not a timestamped field-value history
knowledge reference populated on the joined cohort	0.911	coverage, not timing
base AUC from the knowledge reference alone	0.648	the field is highly predictive, which is a reason for suspicion and not for confidence

Table S23: What the interaction file establishes, and what it does not. Case study cohort, reassignment target.

mechanism	level	populated	V	spread over seeds	seeds
clean	1.000	1.000	0.103	0.000	1
corrupt	0.050	1.000	0.099	0.003	3
corrupt	0.150	1.000	0.086	0.001	3
corrupt	0.300	1.000	0.069	0.002	3
duplicate	0.150	1.000	0.103	0.001	3
duplicate	0.500	1.000	0.104	0.001	3
duplicate	1.000	1.000	0.102	0.001	3
mask_common	0.500	0.521	0.071	0.000	1
mask_random	0.500	0.502	0.068	0.004	3
mask_rare	0.250	0.270	0.041	0.000	1
mask_rare	0.500	0.505	0.082	0.000	1
mask_rare	0.750	0.762	0.096	0.000	1
stale	1.000	0.984	0.103	0.000	1
stale_lag	0.100	0.982	0.107	0.000	1
stale_lag	0.250	0.972	0.106	0.000	1
stale_lag	0.500	0.940	0.107	0.000	1

Table S24: Register-quality mechanisms on the case study's cohort (45,455 incidents) and reassignment target, at the reference cell. 'level' is the share populated for the population mechanisms, the share corrupted for accuracy, the share of identities split for reconciliation and the lag for discovery. Stochastic mechanisms are averaged over 20 seeds and the spread across seeds is printed beside the mean, because mechanism variability and sampling variability are different quantities. The corpus-cohort version of this table is in Supplement S11.

mechanism	levels	seeds	integral	coarse	shift	seed s.d.	V complete	V empty
content	11	3	0.049	0.047	0.001	0.002	0.085	-0.001
mask_rare	11	1	0.069	0.066	0.003	0.000	0.103	-0.003
propensity	11	3	0.056	0.054	0.002	0.002	0.086	-0.001
time	11	1	0.077	0.088	0.011	0.000	0.115	-0.001

Table S25: Register-quality mechanisms as severity CURVES rather than points, on the case study's cohort and reassignment target. The integral is taken against a uniform measure on severity over 11 levels; 'grid shift' is how much coarsening that grid to five levels moves it, which is the check that the summary is a property of the curve and not of the grid.

b lo	b hi	V lo	V hi	D	D lo	D hi	R	R fieller lo	R fieller hi	fieller kind
B_half	B_intake	0.1828	0.1838	-0.0010	-0.0049	0.0033	-0.0056	-0.0270	0.0156	bounded
B_half	B_intake_g	0.1828	0.1034	0.0793	0.0619	0.0896	0.4341	0.3572	0.5127	bounded
B_half	B_intake_g_km	0.1828	-0.0029	0.1857	0.1759	0.2025	1.0157	0.9904	1.0425	bounded
B_intake	B_intake_g	0.1838	0.1034	0.0804	0.0626	0.0914	0.4372	0.3611	0.5151	bounded
B_intake	B_intake_g_km	0.1838	-0.0029	0.1867	0.1773	0.2040	1.0156	0.9905	1.0422	bounded
B_intake_g	B_intake_g_km	0.1034	-0.0029	0.1063	0.0965	0.1308	1.0277	0.9831	1.0771	bounded

Table S26: Absolute absorption $D = V(f \mid B_0) - V(f \mid B_1)$ first, with its own interval, then the ratio R with its Fieller set and the set's kind, on the case study's cohort and reassignment target, in ROC AUC. 0 of 30 reductions across all five instruments have a set that is not a bounded interval; for those a percentile interval is not a summary of anything.

mechanism	level	populated	V
clean	1.000	1.000	0.167
mask_rare	0.750	0.760	0.140
mask_rare	0.500	0.515	0.104
mask_rare	0.250	0.268	0.037
mask_random	0.500	0.502	0.095
mask_common	0.500	0.512	0.106
corrupt	0.050	1.000	0.153
corrupt	0.150	1.000	0.132
corrupt	0.300	1.000	0.097
duplicate	0.150	1.000	0.167
duplicate	0.500	1.000	0.165
duplicate	1.000	1.000	0.163
stale	1.000	0.984	0.167
stale_lag	0.100	0.982	0.169
stale_lag	0.250	0.971	0.169
stale_lag	0.500	0.941	0.164

Table S27: Register-quality mechanisms on the REGISTERED cohort (46,606 cases) and the handover target, at the reference cell, including a discovery lag and a sweep of the two accuracy mechanisms. The case study's own cohort and target are in Table S24.

mechanism	operational failure mode	n levels	n seeds	integral	grid shift	seed s.d.	V complete	V empty
content	discovery: coverage depends on the item's type	11	20	0.090	0.015	0.031	0.146	0.000
corrupt	accuracy: a share of rows carry the wrong identity	11	20	0.113	0.010	0.005	0.068	0.167
duplicate	reconciliation: one item exists under two keys	11	20	0.165	0.001	0.001	0.163	0.167
mask_common	population: the core identities are absent	11	1	0.104	0.018	0.000	0.167	0.014
mask_random	population: absent independently of everything	11	20	0.088	0.016	0.004	0.167	0.000
mask_rare	population: the long tail of identities is absent	11	1	0.092	0.016	0.000	0.167	0.014
propensity	missing where the work is unusual: a training-estimated propensity drives the missingness	11	20	0.069	0.010	0.004	0.119	0.000
time	discovery: the register was populated late	11	1	0.105	0.005	0.000	0.167	0.000

Table S28: Register-quality mechanisms as severity CURVES rather than points, on the primary log at the reference cell. Each row gives the operational failure mode the mechanism represents, the number of severities and seeds, the integral of the increment against a uniform measure on severity, how much coarsening the severity grid by a factor of four moves that integral, the spread across seeds, and the endpoints.

metric	b lo	b hi	V lo	V hi	D	D lo	D hi	R	R fieller lo	R fieller hi	fieller kind
ap	B_half	B_intake	0.031	0.030	0.000	0.000	0.002	0.014	-0.027	0.052	bounded
ap	B_half	B_intake_g	0.031	0.019	0.012	0.009	0.016	0.390	0.276	0.498	bounded
ap	B_half	B_intake_g_km	0.031	0.001	0.030	0.026	0.033	0.976	0.949	1.004	bounded
ap	B_intake	B_intake_g	0.030	0.019	0.012	0.008	0.016	0.381	0.266	0.492	bounded
ap	B_intake	B_intake_g_km	0.030	0.001	0.029	0.026	0.033	0.976	0.948	1.004	bounded
ap	B_intake_g	B_intake_g_km	0.019	0.001	0.018	0.014	0.021	0.961	0.917	1.007	bounded
auc	B_half	B_intake	0.260	0.257	0.003	-0.002	0.015	0.010	-0.021	0.039	bounded
auc	B_half	B_intake_g	0.260	0.167	0.092	0.064	0.127	0.356	0.239	0.469	bounded
auc	B_half	B_intake_g_km	0.260	0.008	0.252	0.229	0.272	0.970	0.946	0.995	bounded
auc	B_intake	B_intake_g	0.257	0.167	0.090	0.063	0.126	0.350	0.229	0.466	bounded
auc	B_intake	B_intake_g_km	0.257	0.008	0.249	0.228	0.271	0.970	0.945	0.995	bounded
auc	B_intake_g	B_intake_g_km	0.167	0.008	0.160	0.122	0.179	0.954	0.917	0.992	bounded
brier_skill	B_half	B_intake	0.269	0.269	0.000	-0.003	0.003	0.001	-0.009	0.012	bounded
brier_skill	B_half	B_intake_g	0.269	0.247	0.022	-0.006	0.049	0.081	-0.026	0.177	bounded
brier_skill	B_half	B_intake_g_km	0.269	0.004	0.265	0.229	0.290	0.985	0.943	1.028	bounded
brier_skill	B_intake	B_intake_g	0.269	0.247	0.021	-0.004	0.049	0.080	-0.029	0.177	bounded
brier_skill	B_intake	B_intake_g_km	0.269	0.004	0.265	0.227	0.292	0.985	0.943	1.028	bounded
brier_skill	B_intake_g	B_intake_g_km	0.247	0.004	0.243	0.208	0.274	0.983	0.938	1.031	bounded
logloss_skill	B_half	B_intake	0.321	0.320	0.001	-0.002	0.002	0.003	-0.004	0.010	bounded
logloss_skill	B_half	B_intake_g	0.321	0.241	0.080	0.060	0.102	0.249	0.188	0.308	bounded
logloss_skill	B_half	B_intake_g_km	0.321	0.015	0.306	0.279	0.328	0.954	0.924	0.985	bounded
logloss_skill	B_intake	B_intake_g	0.320	0.241	0.079	0.060	0.101	0.247	0.186	0.306	bounded
logloss_skill	B_intake	B_intake_g_km	0.320	0.015	0.305	0.279	0.329	0.954	0.924	0.985	bounded
logloss_skill	B_intake_g	B_intake_g_km	0.241	0.015	0.226	0.203	0.244	0.939	0.900	0.980	bounded
nagelkerke	B_half	B_intake	0.363	0.362	0.001	-0.003	0.002	0.003	-0.005	0.010	bounded
nagelkerke	B_half	B_intake_g	0.363	0.266	0.096	0.073	0.121	0.266	0.203	0.326	bounded
nagelkerke	B_half	B_intake_g_km	0.363	0.015	0.348	0.318	0.372	0.959	0.932	0.987	bounded
nagelkerke	B_intake	B_intake_g	0.362	0.266	0.096	0.074	0.121	0.264	0.201	0.324	bounded
nagelkerke	B_intake	B_intake_g_km	0.362	0.015	0.347	0.318	0.372	0.959	0.932	0.987	bounded
nagelkerke	B_intake_g	B_intake_g_km	0.266	0.015	0.252	0.226	0.271	0.944	0.908	0.982	bounded

Table S29: Absolute absorption D first, the ratio R second, with its Fieller set and the set's kind, on the REGISTERED cohort and the handover target, over all five instruments. The case study's own cohort and target are in Table S26.

world	bias vs limit	width (basic)	fixed pct	fixed basic	fixed bc	nested pct	nested basic	nested bc	m-out-of-n
drift	0.000	0.037	0.919	0.917	0.914	0.966	0.919	0.920	–
imbalanced	-0.010	0.116	0.906	0.906	0.900	0.933	0.933	0.923	–
linear	-0.002	0.060	0.925	0.931	0.915	0.939	0.931	0.921	0.768
noisy	-0.002	0.036	0.926	0.924	0.916	0.936	0.948	0.939	–
nonlinear	-0.002	0.044	0.923	0.923	0.924	0.940	0.929	0.929	0.758
sparse	-0.012	0.034	0.631	0.643	0.620	0.372	0.899	0.631	0.728

Table S30: Coverage of the estimator's own limit V_{limit} in each simulated world, under seven interval constructions, at the replicate count of Section 10, with the estimator's bias against that limit and the mean width of the interval this paper reports. Nominal is 0.95; the Monte Carlo standard error on a coverage is 0.7 percentage points, so differences below about a point are not differences. An empty cell is a construction not priced on that world.

n train	K	basic coverage	pct coverage	basic mean width	pct mean width	replicates	coverage se (pp)
2000	20	0.887	0.887	0.085	0.085	150	2.588
2000	200	0.780	0.480	0.042	0.042	150	3.382
2000	1000	0.453	0.100	0.064	0.064	150	4.065
4000	20	0.900	0.907	0.057	0.057	150	2.449
4000	200	0.907	0.500	0.031	0.031	150	2.375
4000	1000	0.507	0.080	0.047	0.047	150	4.082
8000	20	0.893	0.900	0.039	0.039	150	2.520
8000	200	0.893	0.480	0.021	0.021	150	2.520
8000	1000	0.613	0.033	0.034	0.034	150	3.976
8000	3019	0.213	0.000	0.026	0.026	150	3.345

Table S31: Sample size and register cardinality varied jointly. Coverage of V_{limit} and mean interval width for the nested percentile and the nested basic construction. The last row is the cardinality regime of the case study's own register. *How well these are determined.* Every cell is 150 replicates, not the 1,000 of the core experiment, and the coverages here are far from nominal — so the Monte Carlo standard error of a coverage in this table has a median of 3.0 percentage points and reaches 4.1, against the 0.7 that Section 10.3 quotes for the core. It is 3.7 times larger, it is computed at each cell's own observed rate rather than at the nominal one, and it is a column here rather than a sentence because the differences down this table are read row against row.

world	block multiplier	nested basic	nested pct
linear	0.500	0.896	0.944
linear	1.000	0.920	0.908
linear	2.000	0.916	0.932
sparse	0.500	0.856	0.392
sparse	1.000	0.836	0.332
sparse	2.000	0.872	0.408

Table S32: Three block lengths around $n^{1/3}$: the multiplier column is the factor applied to the rule of thumb, so 0.5 halves the block and 2 doubles it. Coverage of V_{limit}.

family	draws are	K	draws	multiplier	mult., MC upper	empirical	emp., OS upper	Rademacher	per-cell
decision-curve	degenerate	534	40	0.392	0.394	0.877	0.889	0.281	0.902
decision-curve	degenerate	1100	80	0.579	0.687	0.892	0.908	0.244	0.922
decision-curve	degenerate	2966	150	0.853	0.854	0.902	0.920	0.821	0.932
decision-curve	gaussian	534	40	0.748	0.756	0.549	0.726	0.601	0.905
decision-curve	gaussian	1100	80	0.859	0.866	0.786	0.923	0.805	0.928
decision-curve	gaussian	2966	150	0.887	0.891	0.846	0.919	0.853	0.937
decision-curve	heavy	534	40	0.760	0.766	0.840	0.894	0.631	0.905
decision-curve	heavy	1100	80	0.838	0.842	0.899	0.920	0.790	0.929
decision-curve	heavy	2966	150	0.854	0.855	0.907	0.923	0.823	0.939
whole-surface	degenerate	120	33	0.835	0.842	0.759	0.888	0.771	0.902
whole-surface	degenerate	180	80	0.927	0.931	0.887	0.974	0.900	0.927
whole-surface	degenerate	480	150	0.936	0.942	0.913	0.969	0.926	0.937
whole-surface	gaussian	120	33	0.824	0.827	0.710	0.821	0.744	0.898
whole-surface	gaussian	180	80	0.900	0.903	0.858	0.955	0.876	0.927
whole-surface	gaussian	480	150	0.909	0.914	0.887	0.948	0.895	0.937
whole-surface	heavy	120	33	0.822	0.827	0.740	0.865	0.749	0.900
whole-surface	heavy	180	80	0.909	0.910	0.878	0.966	0.888	0.924
whole-surface	heavy	480	150	0.928	0.929	0.902	0.967	0.917	0.935

Table S33: Family-wise coverage of the max-t band against a known answer, over synthetic families matched to this corpus in size, draw count, cross-cell correlation and per-cell excess kurtosis. Nominal is 0.95; the five middle columns are the five candidate critical values and the last is the control. *The bootstrap here is ideal* — the draws come from the sampling distribution itself — so every coverage in this table is an upper bound on what the real procedure attains, and the control says why: *per-cell* is the basic interval the band is built from, on the same draws, and it is short of nominal before any multiplicity correction is applied. *Regimes.* `gaussian` is the case the multiplier bootstrap is derived for; `heavy` carries this corpus's measured tails; `degenerate` adds the cells whose net-benefit difference is zero by arithmetic, which the corpus's decision-curve families contain and its surface families do not. Every cell is 2,000 replicates, so no coverage here is determined worse than 1.1%.

world	n train	K	K / n	replicates	coverage at c = 1	se	c	c lo	c hi	c applied	c log-linear
linear	8000	20	0.003	500	0.916	0.012	1.154	1.072	1.288	1.154	1.000
linear	4000	20	0.005	500	0.898	0.014	1.219	1.100	1.314	1.164	1.004
linear	2000	20	0.010	500	0.924	0.012	1.108	1.022	1.228	1.164	1.093
linear	8000	100	0.013	500	0.936	0.011	1.096	0.960	1.205	1.180	1.124
sparse	8000	100	0.013	500	0.892	0.014	1.317	1.221	1.452	1.180	1.124
linear	4000	100	0.025	500	0.896	0.014	1.179	1.087	1.294	1.180	1.223
linear	8000	200	0.025	499	0.932	0.011	1.093	0.985	1.264	1.180	1.223
linear	4000	200	0.050	500	0.888	0.014	1.195	1.119	1.337	1.180	1.331
linear	8000	400	0.050	500	0.884	0.014	1.152	1.106	1.226	1.180	1.331
linear	2000	100	0.050	500	0.912	0.013	1.234	1.053	1.326	1.180	1.331
sparse	4000	200	0.050	500	0.884	0.014	1.182	1.131	1.303	1.180	1.331
linear	4000	400	0.100	499	0.796	0.018	1.287	1.225	1.400	1.314	1.449
linear	2000	200	0.100	500	0.844	0.016	1.316	1.253	1.419	1.314	1.449
sparse	2000	200	0.100	500	0.836	0.017	1.333	1.225	1.372	1.314	1.449
linear	8000	1000	0.125	496	0.655	0.021	1.526	1.418	1.595	1.526	1.490
linear	2000	400	0.200	500	0.758	0.019	1.562	1.418	1.670	1.562	1.578
linear	4000	1000	0.250	500	0.520	0.022	1.739	1.633	1.807	1.727	1.622
sparse	4000	1000	0.250	498	0.460	0.022	1.716	1.672	1.850	1.727	1.622
linear	8000	3019	0.377	488	0.305	0.021	1.880	1.804	1.979	1.876	1.705
linear	2000	1000	0.500	500	0.462	0.022	1.869	1.789	2.024	1.876	1.765

Table S34: The coverage calibration. For each cell of the (n, K) plane: the coverage the reported interval actually attains, and the factor c by which it must be widened to attain the nominal level, with the Monte Carlo interval of that factor. c is the 0.95 quantile of the studentised error and is computed, not searched for. 'c applied' is the monotone regression the corpus is calibrated with; 'c log-linear' is the parametric fit whose slope Section 6.4 quotes, printed beside it because it sits below the measured factor at small ratios and would under-correct there.

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